

## Black Level Auto Adjust Off

The data except this specification conform to those of each product.

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### 1. Product type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX540 series

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX545 series

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series

IMX548-AAMJ-C, IMX548-AAQJ-C

IMX565 series

IMX565-AAMJ-C, IMX565-AAQJ-C, IMX566-AAMJ-C, IMX566-AAQJ-C, IMX567-AAMJ-C, IMX567-AAQJ-C

IMX568 series

IMX568-AAMJ-C, IMX568-AAQJ-C

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### 2. Description

The setting which turn off black level auto adjust function by OB level is described in this document.

In case of long exposure and so on, OB level is offset by leak or any other reason. Because of it, if the adjustment becomes a problem, this setting is introduced for one of the solution.

\* The operation of the sensor is not guaranteed at using this setting. Please use this after evaluating and checking enough. Because Black level auto adjust is turned off, the black level shifts from the value of BLKLEVEL setting. So, we recommend to adjust by ISP or something.

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Application circuits shown, if any, are typical examples illustrating the operation of the devices. Sony Semiconductor Solutions Corporation cannot assume responsibility for any problems arising out of the use of these circuits.

### 3. Features

◆ Black level auto adjust function by OB level is turned off.

◆ Support mode

Normal mode (XVS / XHS sync)

Sequential Trigger mode

Fast Trigger mode

◆ Support readout drive mode

All-pixel scan mode

Vertical / Horizontal 1/2 Subsampling mode

2 × 2 FD binning mode (The product with a color filter is excluded)

ROI mode

Overlap ROI mode

◆ Support interface

(IMX530 series / IMX535 series)

| Interface                                                                            |
|--------------------------------------------------------------------------------------|
| SLVS – EC output (1 lane / 2 lane / 4 lane / 8 lane, 8 bit / 10 bit / 12 bit output) |
| SLVS output ( 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)                           |

(IMX540 series / IMX545 series)

| Interface                                                          |
|--------------------------------------------------------------------|
| SLVS – EC output (1 lane / 2 lane, 8 bit / 10 bit / 12 bit output) |
| SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)   |

(IMX548 series)

| Interface                                                        |
|------------------------------------------------------------------|
| SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output) |

(IMX565 series / IMX568 series)

| Interface                                                                  |
|----------------------------------------------------------------------------|
| CSI-2 serial data output (2 Lane / 4 Lane, 8 bit / 10 bit / 12 bit output) |

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#### 4. Register Map

The register map for Black level auto adjust off is shown below. Set this at Standby mode.  
Please refer to the product specification for register setup other than those list.

Registers corresponding to Chip ID = 07h in Write mode. (I<sup>2</sup>C:35\*\*h)

| Address |                  | Bit | Register Name | Description                                               | Default value after reset |            | Reflection timing |
|---------|------------------|-----|---------------|-----------------------------------------------------------|---------------------------|------------|-------------------|
| 4-wire  | I <sup>2</sup> C |     |               |                                                           | By register               | By address |                   |
| 20h     | 3520h            | 0   |               | 10h: Black level adjust On<br>1Fh: Black level adjust Off | 10h                       | 10h        | S                 |
|         |                  | 1   |               |                                                           |                           |            |                   |
|         |                  | 2   |               |                                                           |                           |            |                   |
|         |                  | 3   |               |                                                           |                           |            |                   |
|         |                  | 4   |               |                                                           |                           |            |                   |
|         |                  | 5   |               |                                                           |                           |            |                   |
|         |                  | 6   |               |                                                           |                           |            |                   |
|         |                  | 7   |               |                                                           |                           |            |                   |
| 40h     | 3540h            | 0   |               | 01h: Black level adjust On<br>00h: Black level adjust Off | 01h                       | 01h        | S                 |
|         |                  | 1   |               |                                                           |                           |            |                   |
|         |                  | 2   |               |                                                           |                           |            |                   |
|         |                  | 3   |               |                                                           |                           |            |                   |
|         |                  | 4   |               |                                                           |                           |            |                   |
|         |                  | 5   |               |                                                           |                           |            |                   |
|         |                  | 6   |               |                                                           |                           |            |                   |
|         |                  | 7   |               |                                                           |                           |            |                   |

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## 5. Note

- When black level automatic adjustment function is turned off, Black level auto adjust doesn't work at changing temperature or leak. So, Black level shifts from the setting value of BLKLEVEL (CID=07h, Address=B4h / B5h). Especially, it is remarkable at high gain.

Please set it properly for use, and use after evaluate and check enough.

- Turn on and turn off this function in the standby state.

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## Revision History

| Version | Date           | Page | Remarks                                                                                                                                                                                                                                                |
|---------|----------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rev.0.1 | 2019 / 4 / 26  | –    | First Edition                                                                                                                                                                                                                                          |
| Rev.1.0 | 2019 / 12 / 9  | –    | First Edition (Official version of IMX530 series / IMX540 series)                                                                                                                                                                                      |
|         |                | 1    | Added: Chapter "Product type"<br><br>Correction: 2 × 2 FD binning mode<br>• 2 × 2 Vertical FD binning mode → 2 × 2 FD binning mode<br>• Added: "The product with a color filter is excluded"<br><br>Added: SLVS output (2 ch) in the support interface |
| Rev.2.0 | 2020 / 12 / 17 | –    | Added: IMX535 series / IMX545 series / IMX565 series                                                                                                                                                                                                   |
|         |                | 2    | Added: CSI-2 serial data output                                                                                                                                                                                                                        |
| Rev.3.0 | 2021 / 3 / 4   | 1, 2 | Added: IMX548 series / IMX568 series                                                                                                                                                                                                                   |
|         |                | 2    | Added: Overlap ROI mode in the Support readout drive mode                                                                                                                                                                                              |

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